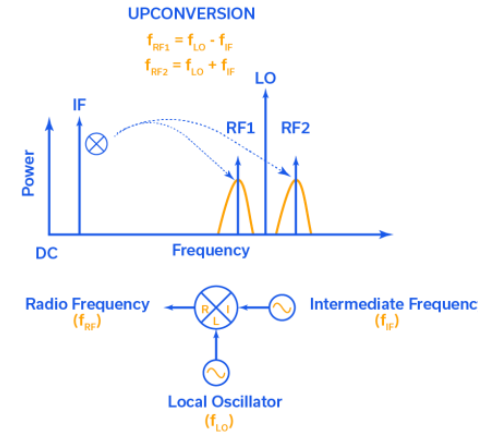
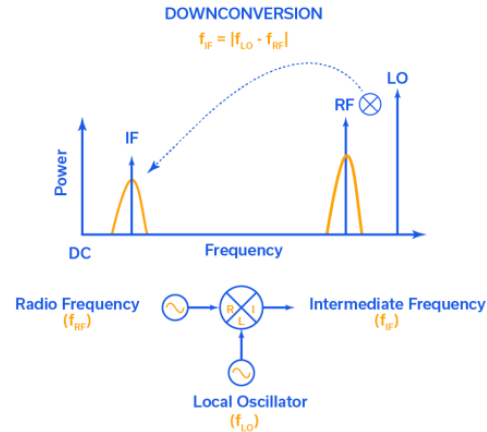

Frequency Conversion Systems

— SAGI Electronics Lecture 8 —

Why Do We Need Frequency Conversion?

- Match a signal to the bandwidth of readout components (down-conversion)
- Condense a wide range of signal frequencies to a single intermediate frequency
- Modulation and Demodulation
- Upconversion in Transmitters
- Frequency Synthesizers
- Spectrum Analyzers



Heterodyne Mixing

- “Mixing” = Multiply two AC signals
- “Heterodyne” = Uses a reference signal (the **local oscillator**) to control another signal

$$V_{LO} = A_{LO} \cos(2\pi f_{LO}t)$$

$$V_{RF} = A_{RF} \cos(2\pi f_{RF}t)$$

$$V_{mix} \propto V_{LO} V_{RF}$$

$$\propto \cos(2\pi f_{LO}t) \cos(2\pi f_{RF}t)$$

$$\propto \cos(2\pi(f_{LO} + f_{RF})t) + \cos(2\pi(f_{LO} - f_{RF})t)$$

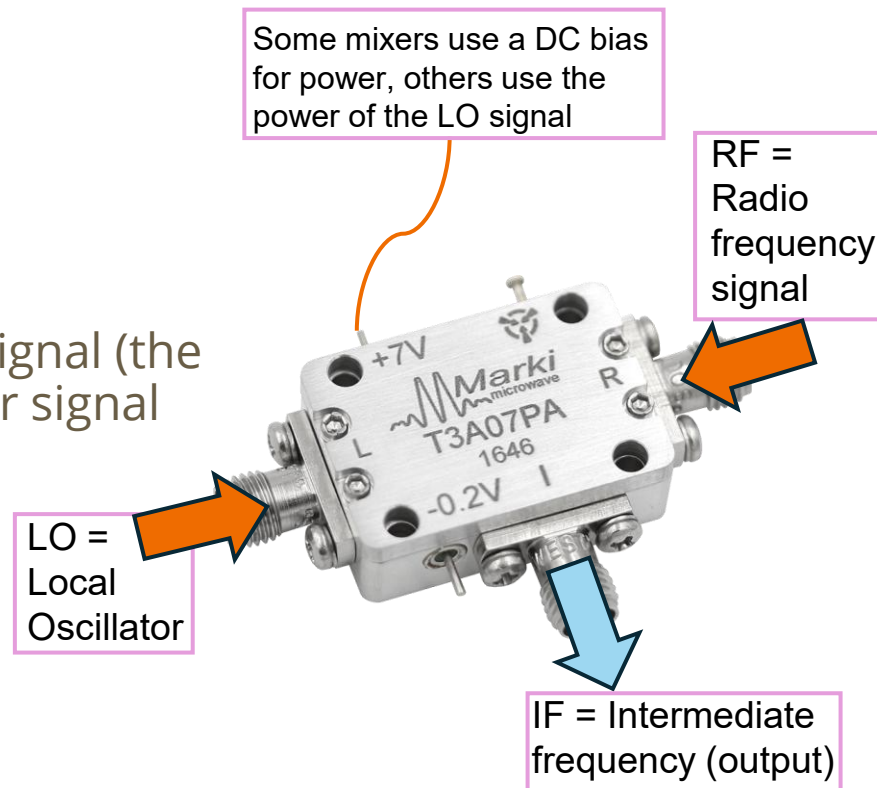


Image Frequency Rejection

- **Image Response:** At the RF port two possible frequencies $f_{in} = f_{LO} \pm f_{IF}$ will be mixed up/down to $f_{out} = f_{IF}$ at the IF output.
- Suppose the RF signal we want to observe is $f_{LO} + f_{IF}$, then we want to reject the signal $f_{LO} - f_{IF}$ before the mixer input.

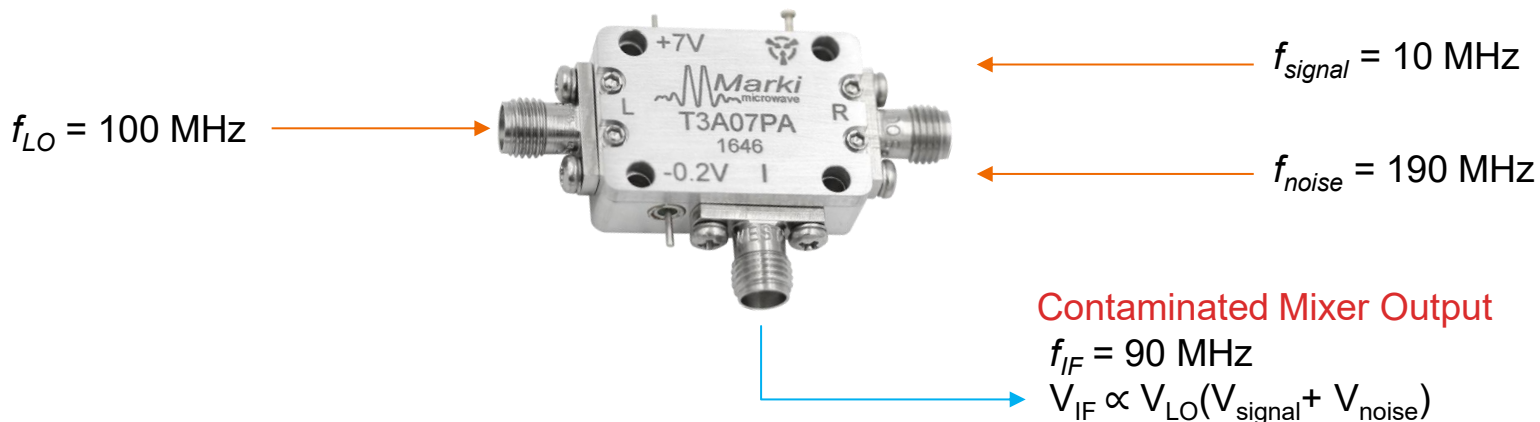
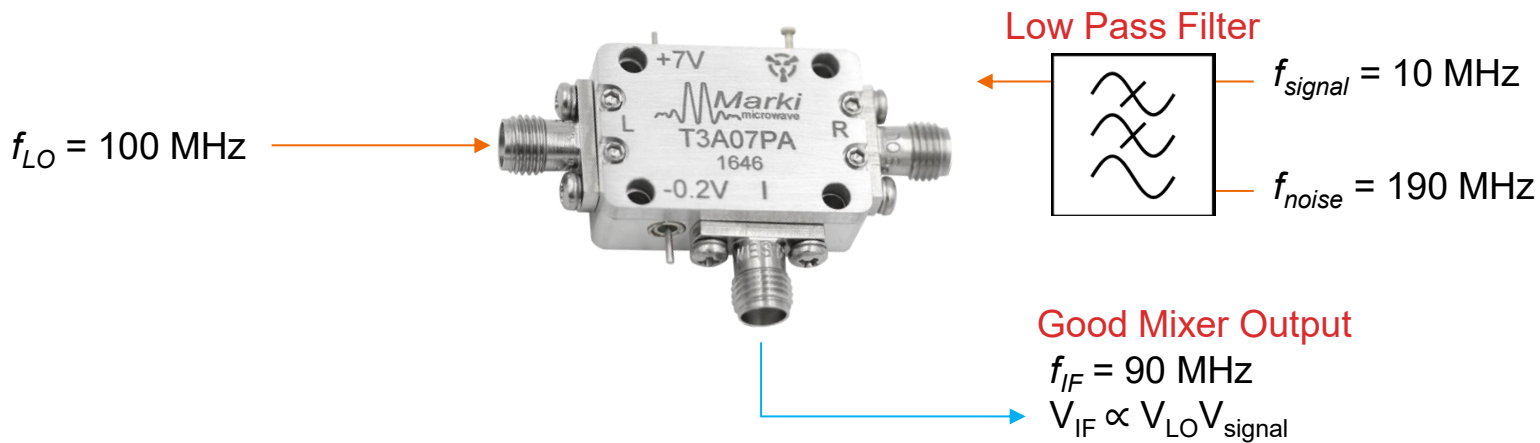


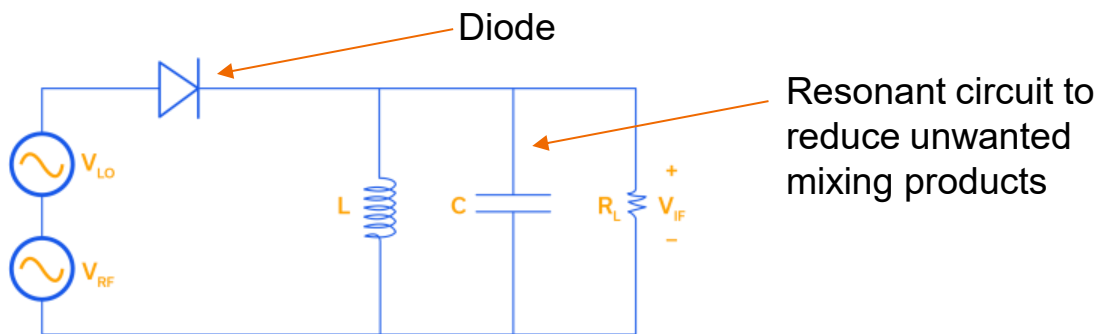
Image Frequency Rejection

- Fix this by putting a filter in front of the RF port to remove any unwanted signals that will get imaged into the IF band.



Passive Mixer Operation

- A **Passive Mixer** is a mixer that does not add gain to the output signal.
- The Single Diode Mixer (Unbalanced Mixer) is the simplest type of mixer.



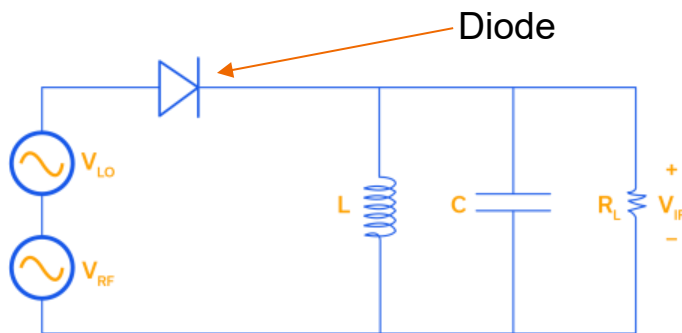
Passive Mixer Operation

- **All mixers** rely on a **non-linear** component for signal transformation.

- Current through diode:

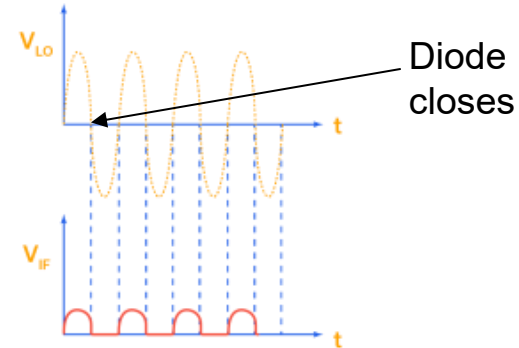
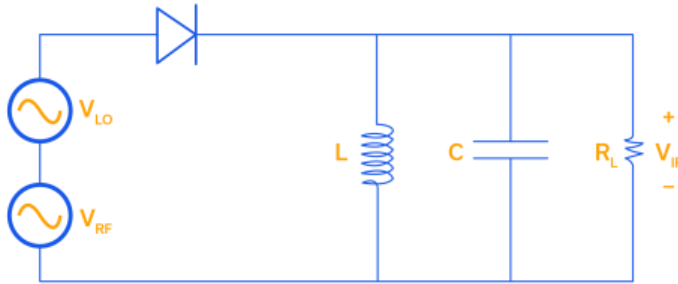
$$I = I_S \left(e^{\frac{qV}{n k_b T}} - 1 \right)$$

- Where I_S is the reverse saturation current of the diode, due to the drift of either electrons or holes across the junction when there is no forward voltage applied
- $1 < n < 2$



Passive Mixer Operation

- The LO signal is applied with much larger voltage than the RF signal.
- This strong LO signal modulates the resistance of the diode, which controls when the combined signal passes through.
- Due to the diode's non-linearity, the current through the diode will contain terms that are products of the input voltages.



Passive Mixer Operation

$$I = I_S \left(e^{\frac{qV}{n k_b T}} - 1 \right)$$

$$I \approx I_S e^{\frac{qV}{n k_b T}}$$

Negligible
with strong
forward bias

- This simple mixer is not very good because it makes a lot of unwanted tones besides just $f_{LO} \pm f_{RF}$

$$V_{in} = V_{RF} + V_{LO}$$

$$V_{in} = A_{RF} \cos(2\pi f_{RF} t) + A_{LO} \cos(2\pi f_{LO} t)$$

$$I \approx I_S \left(\boxed{1 + \alpha V_{in}} + \boxed{\frac{(\alpha V_{in})^2}{2!}} + \boxed{\frac{(\alpha V_{in})^3}{3!} + \frac{(\alpha V_{in})^4}{4!} + \dots} \right)$$

Want to reject

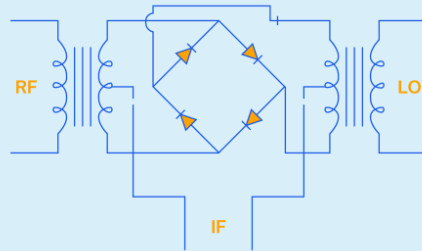
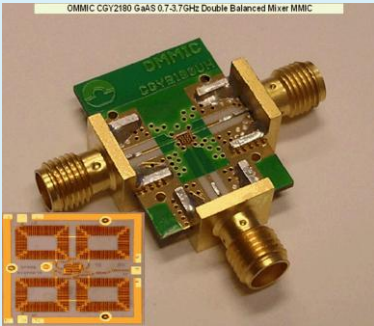
Want to keep because

$$\cos \theta \cos \varphi = \frac{1}{2} (\cos(\theta - \varphi) + \cos(\theta + \varphi))$$

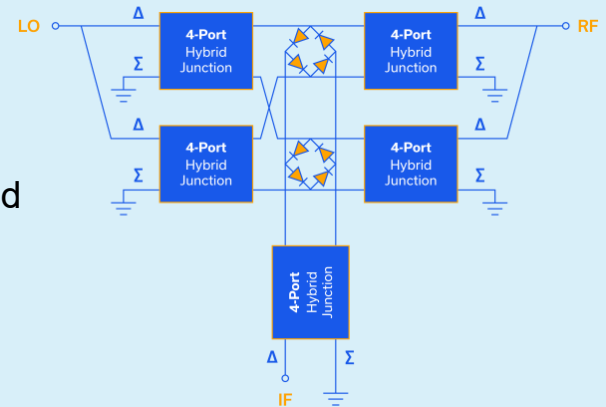
Passive Mixers

- Passive mixers can be engineered to be much more useful and complex than the simple Unbalanced Mixer.
- Balanced = Symmetrical configuration used to cancel unwanted frequencies

Double Balanced Mixer



Double
Balanced
Mixer

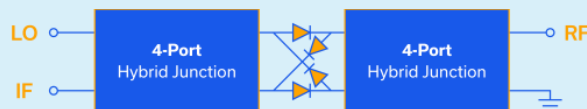


Passive Mixers

- Why use a passive mixer instead of an active mixer?
 - Lower distortion across the bandwidth
 - Does not introduce noise from amplification (only conversion loss)
 - Higher input powers – consistent noise figure at higher powers, circuit can survive at higher power
 - Practicality – Cheaper, less complexity, easy to use

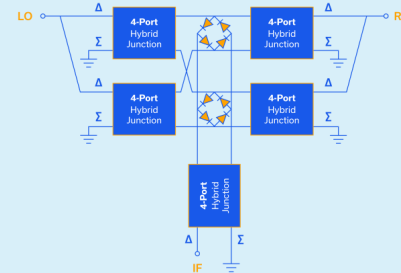
Double Balanced Mixer

- Isolation of RF and LO signals
- Wide band
- No filtering required
- Lower LO power



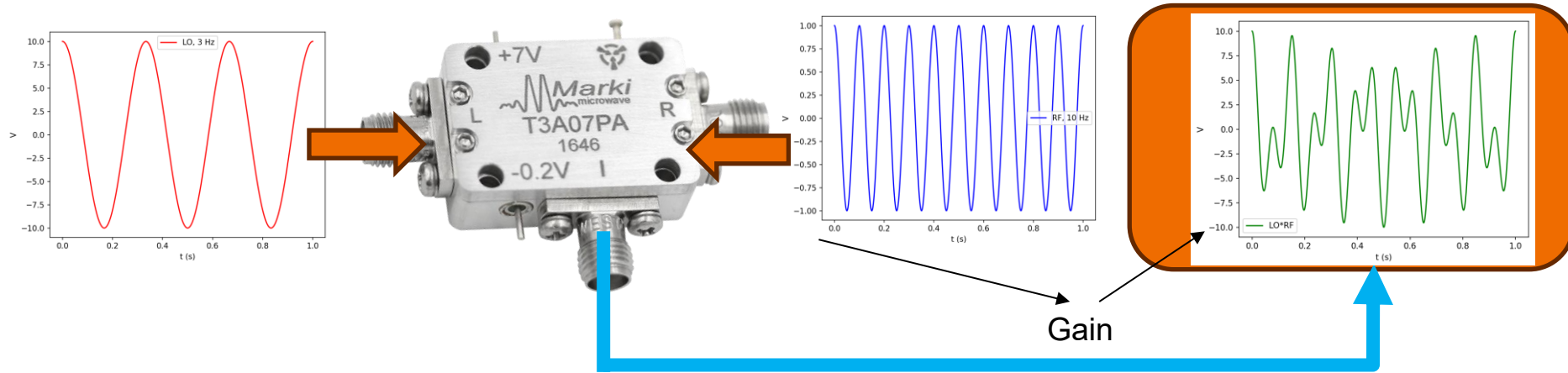
Triple Balanced Mixer

- Best isolation of RF and LO signals
- Wider band
- No filtering required
- High LO power
- Most linear (least distortions)



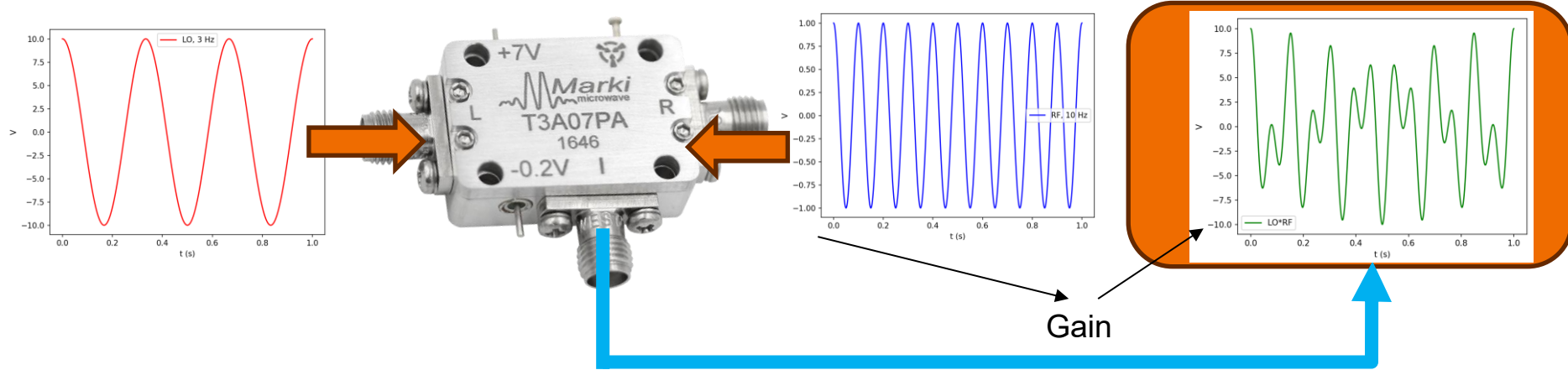
Active Mixer Operation

- An **Active Mixer** is a mixer that does adds gain to the output signal.
- If we took the Fourier Transform of the IF output, what would we see?

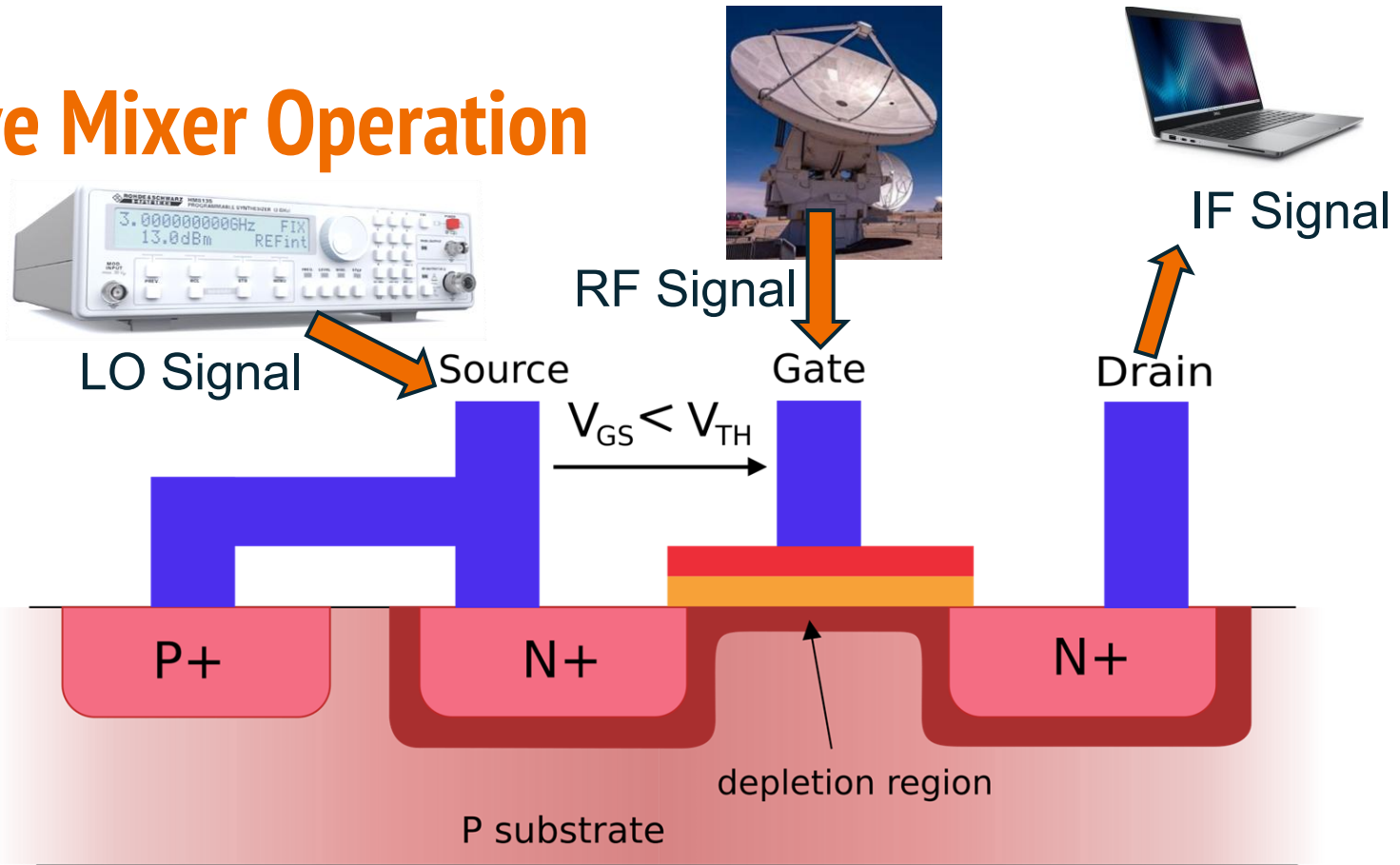


Active Mixer Operation

- Yesterday we learned about RF Power amplifiers and how they add gain. Any guesses how a transistor driven active mixer works?



Active Mixer Operation



Active Mixers

- Why use an active mixer instead of a passive mixer?
 - Can replace both a mixer and an amplifier in the readout chain (“Conversion Gain”)
 - Don’t need to use a high power RF source for the LO
 - Popular in Integrated circuits – cell phones, Wi-Fi cards
 - Better port isolation (no leaks between LO and RF ports)
 - Better Noise Factor at Low RF inputs